BLACK SESAME TECHNOLOGIES

Aura Module Datasheet

NX-5000-D

Data Sheet

Document History

Author	Version	Date	Description of Change
HaoNing	1.0	2026.1.2	Document initial
HaoNing	1.1	2026.2.6	Added : Device ordering information
HaoNing	1.2	2026.5.6	1. Updated and corrected technical parameters. 2. Added Supply power parameter requirements.
HaoNing	1.3	2026.6.29	Updated and corrected technical parameters.
Shichenkang	1.4	2026.6.29	1. Corrected residual typo (Absolute); 2. clarified HMCP lockstep description (6-core Cortex-R5F configured as 3x DCLS); 3. converted full-width punctuation to half-width in English text; 4. removed zero-width characters.
Shichenkang	1.5	2026.6.29	1. Corrected Ethernet port-count wording (2x 2.5GbE + 2x 10GbE); 2. relabeled NNE block (was HiFi DSP).
HaoNing	1.6	2026.7.14	Updated the 314-pin list to align signal usage and carrier-board connection status with the QSC801XP AP-EVB schematic V1.2
Shichenkang	1.7	2026.8.10	Added Product Variant Comparison (NX-5000-D016-M vs NX-5000-D008-L); annotated variant-dependent items in the specifications table; updated Device Ordering Information to the two orderable part numbers.
HaoNing	1.8	2026.9.10	Updated module Storage Temp. Range.
HaoNing Lixuexian	1.9	2026.9.29	1. Update of power domain for module ports, with explanatory notes on port voltages. 2. Add new module system block diagram
HaoNing	1.10	2026.10.10	1. Add the connection requirements for USB0_VBUS on Pin 253. 2. Add the usage notes for the 12-pin connector J1801. ST UART for programming/flashing must used.
HaoNing	1.11	2026.10.21	Update chapter 1.2 "Module Specifications Table", add the performance differences comparison between M and L modules

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Table of Contents

1 Introduction *4*
1.1 Key Advantages 4
1.2 Module Specifications Table 4
2 Pin Definitions *6*
2.1 Module Block Diagram 6
2.2 Pin List 7
2.3 Board to board connector pin definition (J1801) 11
3 Power Supplies *12*
3.1 Supply Current Capabilities & Interface Electrical characteristics 12
3.2 Supply power parameter requirements 12
4 Mechanical Specification *13*
5 Device Ordering Information *14*

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1 Introduction

The SesameX Aura is a high-performance SoC module designed for advanced service robots, quadruped robots, humanoid robots, and autonomous robots.
It integrates multi-core CPU, NPU, GPU, ISP, Codec, Vision DSP, and safety MCU, delivering powerful capabilities in multi-camera perception, AI inference, real-time control, and functional safety, making it well-suited for complex environments requiring multimodal perception and intelligent decision-making.
The Aura module is offered in two variants: NX-5000-D016-M (Mainstream, 16GB LPDDR5) and NX-5000-D008-L (Lite, 8GB LPDDR5). Unless otherwise noted, the specifications in this document apply to both variants; variant differences are summarized in Device Ordering Information.

1.1 Key Advantages

- Powerful AI Performance: Up to 70 TOPS INT8 with sparsity acceleration, supporting tasks such as object detection, semantic segmentation, SLAM
- Multi-camera Perception: Supports 12 channels of 1080p@30fps or 4 channels of 4K@30fps input, with 120dB HDR and night vision enhancement.
- Real-time Control: Heterogeneous MCU cluster with lockstep mode and safety monitoring ensures reliable execution of critical robotic functions.
- High-speed Connectivity: PCIe Gen4, USB 3.1, 2x 2.5GbE + 2x 10GbE with TSN support, providing high-bandwidth connections for sensors and peripherals.
- Safety and Reliability: With automotive-grade safety mechanisms including secure boot, crypto engine, and fault detection.

1.2 Module Specifications Table

	NX-5000-D016-M	**NX-5000-D008-L**
AI Performance (NPU)	**70 TOPS** INT8 (with sparsity)	**65 TOPS** INT8 (with sparsity)
CPU (General-purpose tasks)	Octa-core ARM® Cortex®-A78AE(1.7GHz) • 64KB I- and D- L1 cache • 512KB L2 cache per core • 2MB On-Chip L3 RAM with ECC per Quad-core cluster	
Isolated MPU	Dual-core ARM® Cortex®-A78AE(1GHz)	NA
GPU (visualization multipurpose accelerator)	**Octa**-core ARM® Mali®-G78AE • OpenCL up to 2.2 • OpenGLES 1.1,2.0 and 3.2 • Safety critical drivers for OpenGL SC1.0 and 2.0 • Vulkan 1.0 and 1.1 • Up to 28.8Gtex/s Texture Fill Rate • Up to 14.4Gpix/s Pixel Fill Rate • Up to total 460 GFLOPs(FP32)	**Quad**-core ARM® Mali®-G78AE • OpenCL up to 2.2 • OpenGLES 1.1,2.0 and 3.2 • Safety critical drivers for OpenGL SC1.0 and 2.0 • Vulkan 1.0 and 1.1 • Up to 28.8Gtex/s Texture Fill Rate • Up to 14.4Gpix/s Pixel Fill Rate • Up to total 230 GFLOPs(FP32)
Isolated GPU	Single-core ARM®Mali®-G78AE(900MHz)	NA
HIFI DSP	Single-core NNE(Neural Network Engine)	NA

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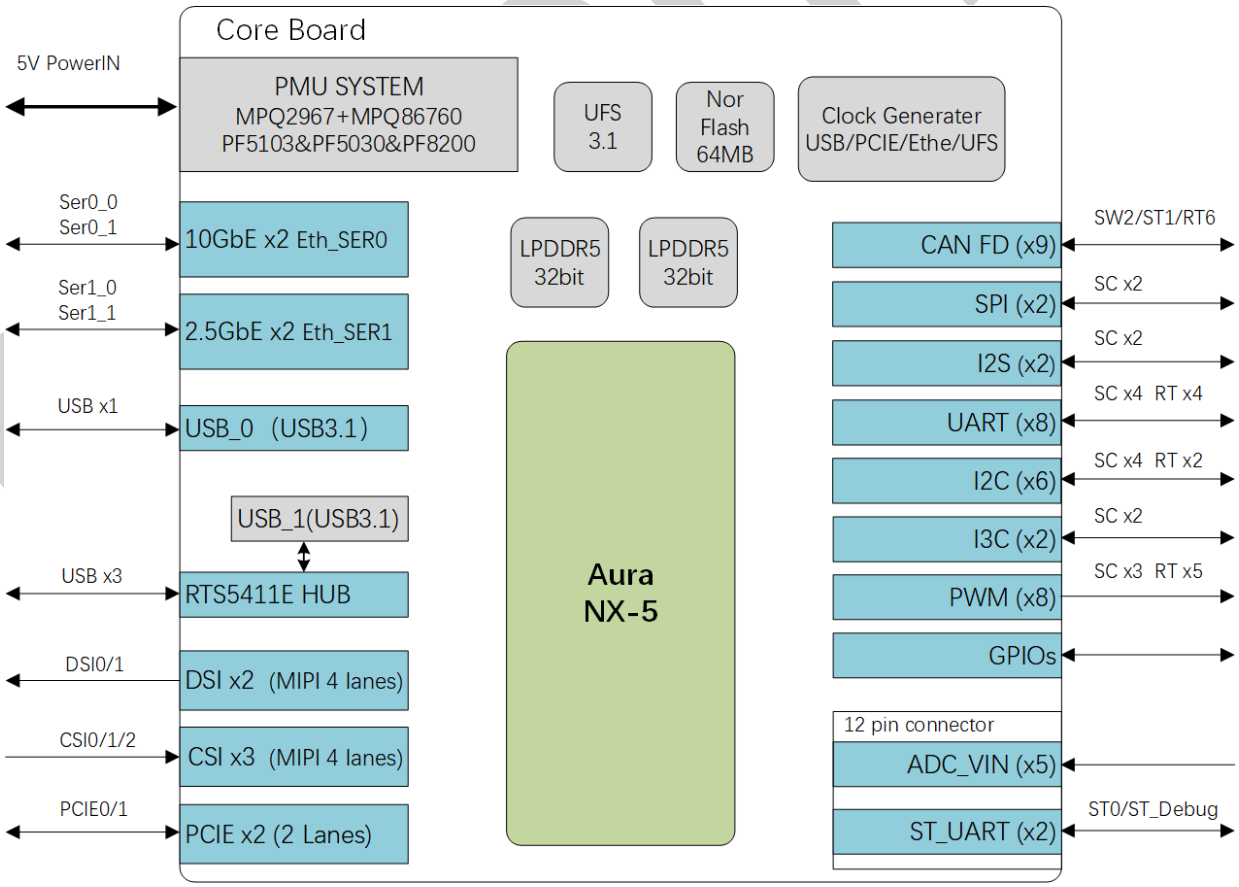
Memory		**2x 8GB** LPDDR5 (16GB) / 64GB UFS3.1	**2x 4GB** LPDDR5 (8GB) / 64GB UFS3.1
Arm Cortex-R5F		• Safety Dual-core ARM Cortex-R5F（1.2GHz） • Security Dual-core ARM Cortex-R5F（1.2GHz） • Realtime Six-core ARM Cortex-R5F(1.2GHz) • Switch Six-core ARM Cortex-R5F(1.2GHz) • DMA R5F Single core ARM® Cortex®-R5F(400MHz)	
ISP		• 4-core ISP@600MHz each core, one-pixel/cycle, supporting three output views (scaling/cropping). • Multi-sensor support: RGGB, RCCB, RYYB, RGB-IR, and 4C sensors. • Image enhancement: lens shading/color correction, defect pixel correction, de-mosaic, sharpening, gamma/contrast/tone mapping. • Advanced features: multi-exposure HDR (up to 3 frames, 120 dB dynamic range), night vision enhancement, LED flicker suppression. • Common operations: auto exposure/white balance, Raw/UV denoise, cropping, resizing, YUV422/420 output.	
Computer Vision DSP		• Dedicated HW accelerators: GWARP (affine/perspective transform with bilinear/bicubic interpolation, RGB/YUV I/O) and Scaler (up/down scaling with polyphase & pyramid modes). • 4 Vision DSPs: each at 1000 MHz, with 64 KB I-Cache & 256 KB DRAM, delivering 256 GMACs (fixed-point) or 32 GFLOPs (float). • Wide SIMD processing: 32-way/64-way parallel computations optimized for CV workloads. • CV function support: APIs for stereo vision, optical flow/SFM, SLAM, image pyramid, and feature extraction (HOG, Harris, ORB).	
Codec	Encode	• Up to 4K@60 • HEVC (H.265): Main, Main10 profiles • H.264: BP, MP, HP profiles • VP8/VP9: VP8, VP9 Profile 0 & Profile 2 (10-bit) • JPEG: Baseline sequential, YUV420/422	
	Decode	• Up to 4K@60 • HEVC (H.265): Main, Main10 profiles • H.264: BP, MP, HP profiles • VP8/VP9 + AVS: VP8, VP9 Profile 0, AVS/AVS2 (Main/Main10) • Others: MPEG4 SP/ASP, MPEG2 MP, H.263 P0, JPEG YUV420/422	
Networking		Ethernet | Media Access Controller (MAC) • 2x 2.5Gbps + 2x10Gbps Support TSN features: • PTP(1588 V2),gPTP(802.1as) • 802.1QAV • 802.1QBV • 802.1CB	
Imaging & Display		3x MIPI CSI Rx Ports • D-PHY(V2.1) Support 4 data-lanes per port,4.5Gbps per lane,up to 12 virtual channels total • C-PHY(V1.2) Support 3 Trio,5.7Gbps per Trio	

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	2x MIPI DSI TX Ports • D-PHY(V1.2) Support 4 data-lanes per port,2.5Gbps per lane
External Storage	Nor flash 512Mb
USB	4 x USB 3.X (3 x USB3.0 host only, 1 x USB3.1 host&device),compatibility with USB 2.0
PCIe	• Support RC or EP mode • Configurable to 2x 2lane or 1x 4lane mode • Support max speed Gen4(16GT/s),compliant with Gen3(8GT/s),Gen2(5GT/s),Gen1(2.5GT/s)
Universal Peripherals	• 9xCANFD/ 2xSPI/ 8xPWM/ 2xI2S/ 6xI2C/ 2xI3C/ 8xUART/ 5xADC/ GPIOs
Mechanical	• 82mm*54mm | 314 Pin MXM + 12 Pin B2B
Operating Requirements	• Operating Temp. Range : -25~70℃ • Storage Temp. Range : -40~85℃ • ESD(HBM) : < 2KV

2 Pin Definitions

2.1 Module Block Diagram



2.2 Pin List

Pin NO. Odd	Module Function Net Name	SOC Description &Usage	Type	Aura Carrier Board	Pin NO. Even	Module Function Net Name	SOC Description &Usage	Type	Aura Carrier Board
1	VCC_SOC_5V0	Module Main 5V power supply	Power Input		2	VCC_SOC_5V0	Module Main 5V power supply	Power Input	
3	VCC_SOC_5V0				4	VCC_SOC_5V0			
5	VCC_SOC_5V0				6	VCC_SOC_5V0			
7	VCC_SOC_5V0				8	VCC_SOC_5V0			
9	VCC_SOC_5V0				10	VCC_SOC_5V0			
11	VCC_SOC_5V0				12	VCC_SOC_5V0			
13	VCC_SOC_5V0				14	VCC_SOC_5V0			
15	VCC_SOC_5V0				16	VCC_SOC_5V0			
17	VCC_SOC_5V0				18	VCC_SOC_5V0			
19	GND	GND			20	GND	GND		
21	GND				22	GND			
23	GND				24	GND			
25	GND				26	GND			
27	GND				28	GND			
29	GND				30	GND			
31	GND				32	GND			
33	GND				34	GND			
35	GND				36	GND			
37	SC_I2S0_SCK	SC_I2S0 Bus Port	1.8V	30 Pin J0902	38	100OHM_MIPI_DSI0_CSITX_D3_N	Dsi0 mipi data lane 0~3		Y HDMI IC
39	SC_I2S0_MCK				40	100OHM_MIPI_DSI0_CSITX_D3_P			
41	SC_I2S0_DIN0				42	100OHM_MIPI_DSI0_CSITX_D2_N			
43	SC_I2S0_WS				44	100OHM_MIPI_DSI0_CSITX_D2_P			
45	SC_I2S0_DOUT0				46	100OHM_MIPI_DSI0_CSITX_D1_N			
47	SC_I2S1_DIN	SC_I2S1 Bus Port		Y HDMI IC	48	100OHM_MIPI_DSI0_CSITX_D1_P			
49	SC_I2S1_WS				50	100OHM_MIPI_DSI0_CSITX_D0_N			
51	SC_I2S1_DOUT				52	100OHM_MIPI_DSI0_CSITX_D0_P			
53	SC_I2S1_MCK				54	GND			
55	SC_I2S1_SCK				56	100OHM_MIPI_DSI0_CSITX_CK_N	Dsi0 mipi Clock		Y HDMI IC
57	GND				58	100OHM_MIPI_DSI0_CSITX_CK_P			
59	VDDIO_AON_1V8	From PF5103	Power out 1.8V		60	GND			
61	VDDIO_ST_IF_1V8	From MPF5030			62	100OHM_MIPI_DSI1_D3_N	Dsi1 mipi data lane 0~3		Y J0702
63	VDDIO_SOC_1V8	From PF8200			64	100OHM_MIPI_DSI1_D3_P			
65	VDDIO_SDIO_1	From PF8200	1.8/3.0V Default 3.0V		66	100OHM_MIPI_DSI1_D2_N			
67	GND				68	100OHM_MIPI_DSI1_D2_P			
69	SOC_RESET	SOC_PGOOD reset	Input 1.8V	Y	70	100OHM_MIPI_DSI1_D1_N			
71	SC_SPI_M0_SDI	SC_SPI_M0 Bus Port	1.8V	30 Pin J0902	72	100OHM_MIPI_DSI1_D1_P			
73	SC_SPI_M0_SDO				74	100OHM_MIPI_DSI1_D0_N			
75	SC_SPI_M0_SCLK				76	100OHM_MIPI_DSI1_D0_P			
77	SC_SPI_M0_CS				78	GND			
79	SC_SPI_M1_SDO	SC_SPI_M1 Bus Port		26 Pin J0801	80	100OHM_MIPI_DSI1_CK_N	Dsi1 mipi Clock		Y
81	SC_SPI_M1_SCLK				82	100OHM_MIPI_DSI1_CK_P			
83	SC_SPI_M1_CS				84	GND			

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85	SC_SPI_M1_SDI				86	RT_PWM_OUT8	Carrier board PWM	1.8V	
87	SC_UART3_CTS	SC_UART3 Bus Port		30 Pin J0902	88	RT_I2C0_SDA_PWM7	Carrier board PWM_OUT7	1.8V	20 Pin J0901
89	SC_UART3_TXD				90	RT_I2C0_SCL_PWM6	Carrier board PWM_OUT6		
91	SC_UART3_RXD				92	RT_PWM_OUT13	Carrier board PWM_OUT13		TP2133
93	SC_UART3_RTS				94	RT_PWM_OUT12	Carrier board PWM_OUT5		20 Pin
95	SC_UART1_RXD	SC_UART1 Bus Port		Y	96	RT_PWM_OUT15	From AL13 pin	1.8V	Float
97	SC_UART1_TXD				98	RT_PWM_OUT14	From AL14 pin	1.8V	Float
99	SC_UART0_RXD	SC_UART0 Bus Port		Y	100	SW_FRAY0_A_RXDSC_PWM0	Carrier board FAN PWM Ctrl	J0401 1.8V	Y
101	SC_UART0_TXD				102	GND			
103	SC_UART2_TXD	SC_UART2 Bus Port		Y	104	RT_UART3_RTS	PCIE M.2 key E UART	1.8V	Y
105	SC_UART2_RXD				106	RT_UART3_CTS			
107	GND				108	RT_UART3_TXD			
109	SC_I2C4_SDA	SC_I2C4 Bus Port	1.8V	30 Pin J0902	110	RT_UART3_RXD			
111	SC_I2C4_SCL				112	RT_UART0_TXD	RT UART Ports		20 Pin J0901
113	SC_I2C5_SDA	SC_I2C5 Bus Port			114	RT_UART0_RXD			
115	SC_I2C5_SCL				116	RT_UART1_TXD			
117	SC_I2C6_SCL	SC_I2C6 Bus Port			118	RT_UART1_RXD			
119	SC_I2C6_SDA				120	RT_UART2_TXD			
121	RT_I2C2_SCL	RT_I2C2 Bus Port		Y	122	RT_UART2_RXD			
123	RT_I2C2_SDA				124	PMIC_PWRON	module power ready feedback	IN 5V	Y
125	RT_I2C3_SCL	RT_I2C3 Bus Port		Y HDMI IC	126	ST_RGMII_PPSO0		1.8V	Y
127	RT_I2C3_SDA				128	GND			
129	SC_I2C7_SDA	SC_I2C7 Bus Port		Y	130	SW_CANFD0_TX	SW_CANFD Ports	1.8V	20 Pin J0901
131	SC_I2C7_SCL				132	SW_CANFD0_RX			
133	SC_I3C0_SCL	SC_I3C0 Bus Port		Float	134	SW_CANFD1_TX			
135	SC_I3C0_SDA0				136	SW_CANFD1_RX			
137	SC_I3C1_SCL	SC_I3C1 Bus Port		Float	138	ST_CANFD0_TX	ST_CANFD Ports		26 Pin J0801
139	SC_I3C1_SDA				140	ST_CANFD0_RX			
141	GND				142	RT_CANFD0_TX	RT_CANFD Ports		26 Pin J0801
143	RT_CANFD5_RX	RT_CANFD Ports	1.8V	Float	144	RT_CANFD0_RX			
145	RT_CANFD5_TX				146	GND			
147	RT_CANFD4_RX				148	WK_GPIO35	CSI2 MIPI Mclk	Out 1.8V	Y
149	RT_CANFD4_TX				150	WK_GPIO34	CSI1 MIPI Mclk	Out 1.8V	
151	RT_CANFD3_RX			Pin4 J1402	152	WK_GPIO33	WK GPIO	Out 1.8V	
153	RT_CANFD3_TX			Pin4 J1401	154	WK_GPIO28	M.2 key E BT_WAKE_HOST	Input 1.8V	
155	RT_CANFD2_RX	RT_CANFD Ports		26 Pin J0801	156	WK_GPIO29	HDMI INT	Out 1.8V	
157	RT_CANFD2_TX				158	WK_GPIO32	Wake up Key	Input 1.8V	
159	RT_CANFD1_RX				160	WK_GPIO36	IO_Expand_INT4 to host	Input 1.8V	
161	RT_CANFD1_TX				162	WK_GPIO37	Type-C OTG flag	Input	

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								1.8V	
163	GND				164	WK_PWR_EN	Power on delay control	Out 1.8V	Y
165	SC_PCM3_IN	SC_PCM3 Bus Ports		Y		KEY			
	KEY								
167	SC_PCM3_OUT	SC_PCM3 Bus Ports	1.8V	Y J1901	166	SC_GPIO160	IO_Expand_USB0_OVP	Out 1.8V	Y
169	SC_PCM3_CLK				168	SC_GPIO159	IO_Expand_INT3 to host	Input 1.8V	
171	SC_PCM3_SYNC				170	WK_GPIO11	SW_CANFD0_STB	Out 1.8V	
173	50OHM_SC_SD_DAT1	SDMMC Bus Ports	1.8/3.0V, Default 3.0V (VDDIO_SDIO_1)	Float	172	SC_GPIO154	AW35615 INT_N	Input 1.8V	Y
175	50OHM_SC_SD_DAT0				174	SC_GPIO149	SW_CANFD0_EN	Out 1.8V	Y
177	50OHM_SC_SD_CMD				176	GND			
179	50OHM_SC_SD_CLK				178	85OHM_ETH_SER0_TX0_P	XGMAC Ser0 transmit and receive lanes		Y
181	50OHM_SC_SD_RCLK				180	85OHM_ETH_SER0_TX0_N			
183	SC_SDMMC0_WP				182	85OHM_ETH_SER0_TX1_P			
185	GND				184	85OHM_ETH_SER0_TX1_N			
187	SW_XGMAC2_MDC	XGMAC Control Ports	1.8V	Y	186	85OHM_ETH_SER0_RX0_P			
189	SW_XGMAC2_TRIG_IN1			Float	188	85OHM_ETH_SER0_RX0_N			
191	SW_XGMAC0_MDC			Y	190	85OHM_ETH_SER0_RX1_P			
193	SW_XGMAC2_MDIO			Y	192	85OHM_ETH_SER0_RX1_N			
195	SW_XGMAC2_PPSO0			Y	194	GND			
197	SW_XGMAC0_MDIO			Y	196	85OHM_PCIE1_TX0_P	PCIE1 Port lanes M.2 key M		Y J1702
199	SW_XGMAC2_TRIG_IN0			Float	198	85OHM_PCIE1_TX0_N			
201	SW_XGMAC2_PPSO1			Y	200	85OHM_PCIE1_TX1_P			
203	GND				202	85OHM_PCIE1_TX1_N			
205	85OHM_ETH_SER1_TX1_P	XGMAC Ser1 Dual transmit and Dual receive lanes		Float	204	85OHM_PCIE1_RX0_P			
207	85OHM_ETH_SER1_TX1_N				206	85OHM_PCIE1_RX0_N			
209	85OHM_ETH_SER1_TX0_P				208	85OHM_PCIE1_RX1_P			
211	85OHM_ETH_SER1_TX0_N				210	85OHM_PCIE1_RX1_N			
213	85OHM_ETH_SER1_RX0_P				212	GND			
215	85OHM_ETH_SER1_RX0_N				214	85OHM_PCIE1_CLK_P	PCIE1 clock M.2 key M		Y J1702
217	85OHM_ETH_SER1_RX1_P				216	85OHM_PCIE1_CLK_N			
219	85OHM_ETH_SER1_RX1_N				218	GND			
221	GND				220	100OHM_USB0_TX1_N	USB 3.1 Type-C Data transmit and receive lanes from SOC		USB-C J1501
223	85OHM_PCIE0_TX0_P	PCIE0 TX0 lanes M.2 key E		Y J1901	222	100OHM_USB0_TX1_P			
225	85OHM_PCIE0_TX0_N				224	100OHM_USB0_TX0_N			
227	85OHM_PCIE0_TX1_P	PCIE0 Port TX1 lanes		Float	226	100OHM_USB0_TX0_P			
229	85OHM_PCIE0_TX1_N				228	100OHM_USB0_RX1_N			
231	85OHM_PCIE0_RX0_P	PCIE0 TX0 lanes M.2 key E		Y J1901	230	100OHM_USB0_RX1_P			
233	85OHM_PCIE0_RX0_N				232	100OHM_USB0_RX0_N			
235	85OHM_PCIE0_RX1_P	PCIE0 Port RX1 lanes		Float	234	100OHM_USB0_RX0_P			
237	85OHM_PCIE0_RX1_N				236	GND			
239	GND				238	90OHM_USB0_DN	USB0 2.0 Type-C Data lane		USB-C J1501
241	85OHM_PCIE_X4_REF_CLK_P	PCIE0 clock M.2 key E		Y J1901	240	90OHM_USB0_DP			
243	85OHM_PCIE_X4_REF_CLK_N				242	GND			

Design A

245	GND				244	100OHM_MIPI_CSI0_RX_D3P			
247	PCIE1_PERSTN	PCIE Control Port	1.8V	Y	246	100OHM_MIPI_CSI0_RX_D3N			
249	PCIE0_PERSTN		1.8V	Y	248	100OHM_MIPI_CSI0_RX_D2P			
251	USB0_ID	SOC Not support OTG *1		Float	250	100OHM_MIPI_CSI0_RX_D2N			
253	USB0_VBUS	A 30K±% resistor needs to be connected through this pin to VBUS. Should not be directly connected to the 5V VBUS .	5V	Y	252	100OHM_MIPI_CSI0_RX_D1P	CSI0 mipi data lane 0~3		Float
255	GND				254	100OHM_MIPI_CSI0_RX_D1N			
257	100OHM_DSP1_SS_TXP	USB 3.0 Type-C Data transmit and receive lanes from usb hub		USB-C J1803	256	100OHM_MIPI_CSI0_RX_D0P			
259	100OHM_DSP1_SS_TXM				258	100OHM_MIPI_CSI0_RX_D0N			
261	100OHM_DSP1_SS_RXP				260	GND			
263	100OHM_DSP1_SS_RXM				262	100OHM_MIPI_CSI0_RX_CKP	CSI0 mipi clock lane		Float
265	GND				264	100OHM_MIPI_CSI0_RX_CKN			
267	90OHM_DSP1_DP	USB 2.0 Type-C Data lanes		USB-C J1803	266	GND			
269	90OHM_DSP1_DM				268	100OHM_MIPI_CSI1_RX_D0N	CSI1 mipi data lane 0~3		Y J1401
271	GND				270	100OHM_MIPI_CSI1_RX_D0P			
273	90OHM_DSP2_DP	USB 2.0 Type-A1 Data lanes		USB-A J1802	272	100OHM_MIPI_CSI1_RX_D1N			
275	90OHM_DSP2_DM				274	100OHM_MIPI_CSI1_RX_D1P			
277	GND				276	100OHM_MIPI_CSI1_RX_D2N			
279	100OHM_DSP2_SS_TXP	USB 3.0 Type-A1 Data transmit and receive lanes		USB-A J1802	278	100OHM_MIPI_CSI1_RX_D2P			
281	100OHM_DSP2_SS_TXM				280	100OHM_MIPI_CSI1_RX_D3N			
283	100OHM_DSP2_SS_RXP				282	100OHM_MIPI_CSI1_RX_D3P			
285	100OHM_DSP2_SS_RXM				284	GND			
287	GND				286	100OHM_MIPI_CSI1_RX_CKN	CSI1 mipi clock lane		Y J1401
289	100OHM_DSP3_SS_TXP	USB 3.0 Type-A2 Data transmit and receive lanes		USB-A J1802	288	100OHM_MIPI_CSI1_RX_CKP			
291	100OHM_DSP3_SS_TXM				290	GND			
293	100OHM_DSP3_SS_RXP				292	100OHM_MIPI_CSI2_RX_D0N	CSI2 mipi data lane 0~3		Y J1402
295	100OHM_DSP3_SS_RXM				294	100OHM_MIPI_CSI2_RX_D0P			
297	GND				296	100OHM_MIPI_CSI2_RX_D1N			
299	90OHM_DSP3_DP	USB 2.0 Type-A2 Data lanes		USB-A J1802	298	100OHM_MIPI_CSI2_RX_D1P			
301	90OHM_DSP3_DM				300	100OHM_MIPI_CSI2_RX_D2N			
303	GND				302	100OHM_MIPI_CSI2_RX_D2P			
305	OCP3	USB HUB A2 OCP detect	3.3V	Y	304	100OHM_MIPI_CSI2_RX_D3N			
307	OCP2	USB HUB A1 OCP detect	3.3V	Y	306	100OHM_MIPI_CSI2_RX_D3P			
309	OCP1	USB HUB C OCP detect	3.3V	Y	308	GND			
311	DSP3_PWR	from USB HUB en USB A2 Vbus		Y	310	100OHM_MIPI_CSI2_RX_CKN	CSI2 mipi clock lane		Y J1402
313	DSP2_PWR	from USB HUB en USB A1 Vbus		Y	312	100OHM_MIPI_CSI2_RX_CKP			
314	DSP1_PWR	from USB HUB en J1803 C Vbus		Y					

Notes:

Y	Connect to Carrier Board IC as function nets
Float	Only Golden Finger On Module core board
30 Pin	Through voltage converter to Carrier Board 30pin header J0902
26 Pin	Through voltage converter to Carrier Board 26pin header J0801
20 Pin	Through voltage converter to Carrier Board 20pin header J0901

*1 :Type-C DRP by external CC controller, not dependent on USB0_ID

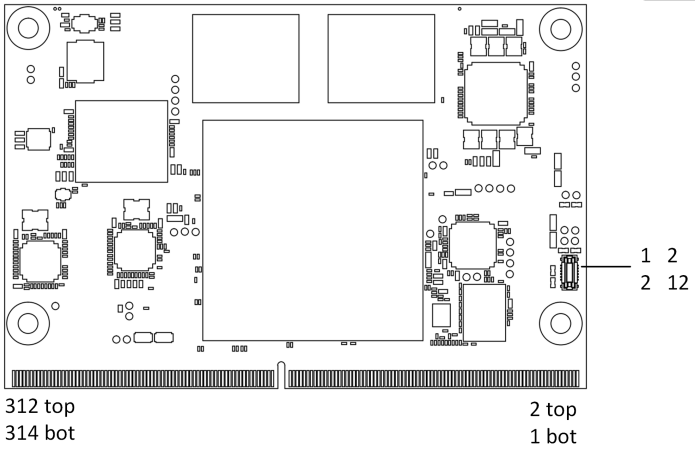
Seeed

2.3 Board to board connector pin definition (J1801)

1	**ST_DBUART0_TXD**	2	ADC_VINS0
3	**ST_DBUART0_RXD**	4	ADC_VINS1
5	GND	6	ADC_VINS2
7	**ST_DBUART_TXD**	8	ADC_VINS3
9	**ST_DBUART_RXD**	10	ADC_VINS4
11	GND	12	GND

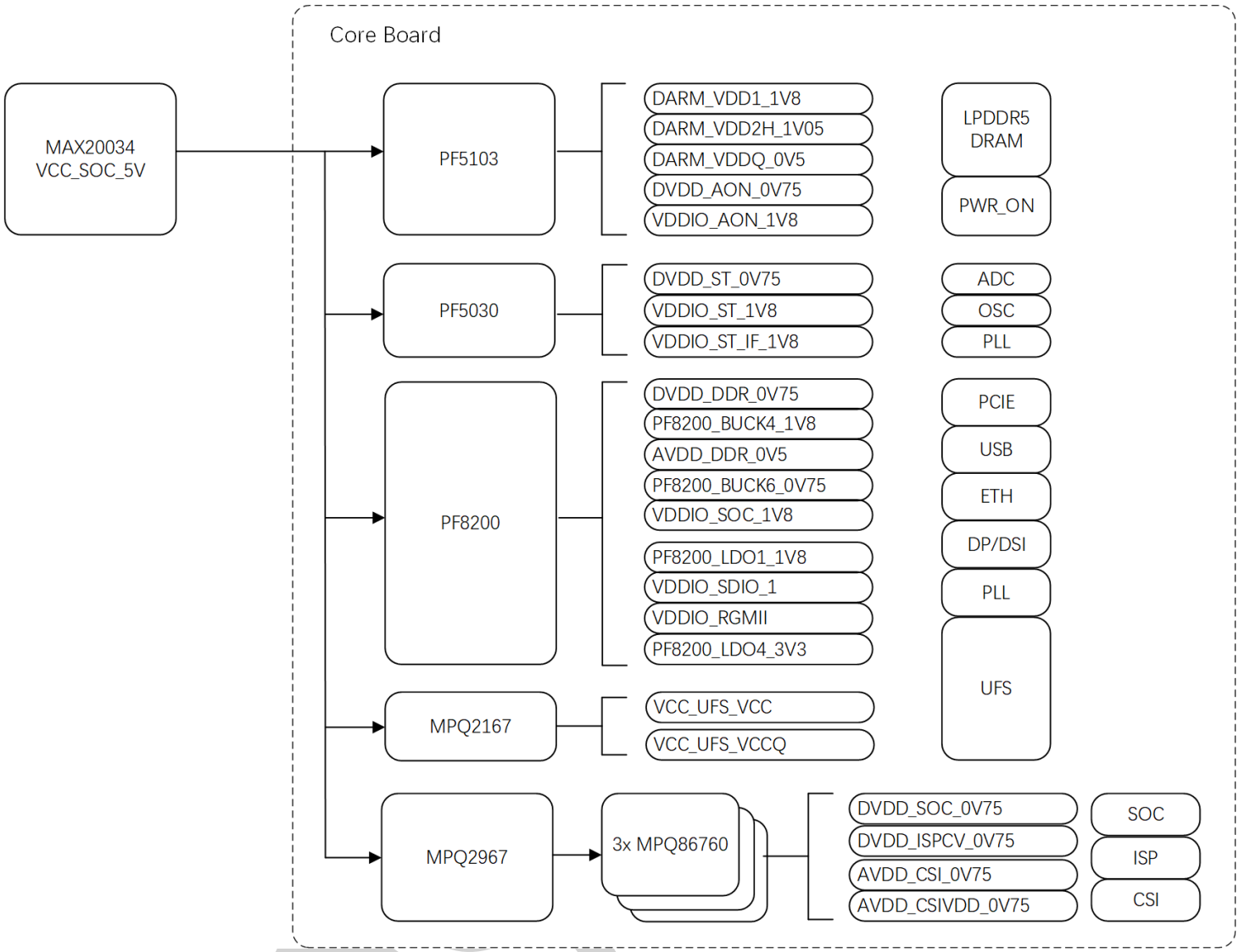
Notes:

ST_DBUART0 and ST_DBUART （J1801 PIN 1/3/7/9） must be connected externally for the purpose of flashing the firmware configuration.



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3 Power Supplies



3.1 Supply Current Capabilities & Interface Electrical characteristics

Descriptions	Usage & Characteristics	Vmin	V	Vmax	Current (A)
VCC_SOC_5V0	Core board 5V Power supply Input	4.75	5	5.25	
GPIO	Absolute input & output voltage	0	1.8	1.98	0.2
SD/EMMC IO	Absolute input & output voltage	0	3	3.3	0.2

3.2 Supply power parameter requirements

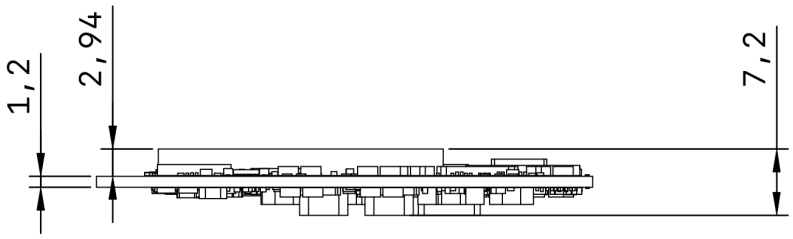
VCC_SOC_5V0 serves as the power input supply for the core board and shall meet the following power input requirements. The test point shall be located at the core board connector. Anti-surge protection shall also be implemented.

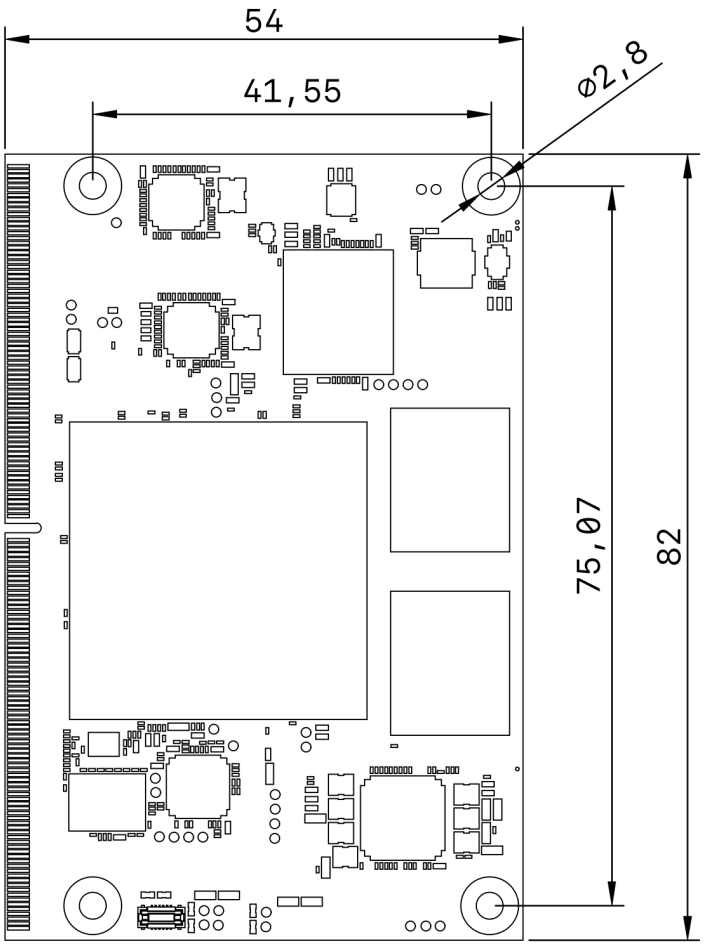
Descriptions	Parameter Requirements
Nominal Input Voltage（标称电压）	5.0 V
Operating Voltage Range（稳态工作范围）	4.75 V ~ 5.25 V

Absolute DC Maximum Voltage at PMIC Input （PMIC 输入端绝对直流上限）	≤5.5 V
Ripple and Noise（纹波噪声）	< 150 mVpp,20 MHz BW
Transient Voltage Droop（瞬态跌落）	≥4.65 V
Transient Voltage Overshoot（瞬态过冲）	≤5.35 V,Absolute < 5.5 V
Recommended External Power Supply（推荐电源）	5 V / 20 A ,minimum ≥13A
Peak Current Capability（峰值电流能力）	≥25 A
Overvoltage Protection （OVP 硬保护）	5.40 V ~ 5.50 V
Undervoltage Protection （UVP 保护）	4.50 V ~ 4.65 V
Anti-surge protection（反灌保护）	必须考虑

4 Mechanical Specification

Unit: mm





5 Device Ordering Information

NX-5000-D008-L	Memory Config: 8GB DRAM	L : Lite
NX-5000-D016-M	Memory Config: 16GB DRAM	M : Mainstream

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